ClozeCongruence: Model Fingerprinting and Traceback Attribution in Large Language Models via Multi-Density Reverse Cloze Resonance and Micro-Stylometric DNA

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Abstract *— As large language models (LLMs) proliferate across academia, journalism, legal discovery, and cyber forensics, identifying not merely whether a text was machine-generated, but which specific foundation architecture authored it, has emerged as a paramount challenge. Existing attribution approaches depend on fragile statistical perplexity thresholds, server-side watermarks requiring generation-time intervention, or white-box logit extraction inaccessible under proprietary commercial APIs. We present* ***ClozeCongruence****, a non-invasive, training-free, and model-agnostic provenance traceback framework founded upon two core phenomena:* ***Self-Model Cloze Resonance*** *and the* ***Reverse Cloze Convergence Gradient (Δ)****, integrated with a* ***15-dimensional Micro-Stylometric DNA Vector****. When candidate LLMs reconstruct cloze-masked spans of an unknown text, the true authoring architecture exhibits a massive* ***Self-Model Resonance Gap (+82.21% mean advantage, $p < 10^{-14}$)*** *over non-authoring models. Transitioning from high-density centroid 3-sentence masking (Pass 3) to low-density alternating masking (Pass 2) triggers a strictly positive convergence gradient (Δ* ***= +11.50%****) for the authentic model, whereas non-authoring models remain stagnant (Δ ≈ 0%). To resolve multi-sentence blocks, we formalize an exact combinatorial assignment over the symmetric group S_3 solvable in O(1) constant time (<0.05 ms), coupled with continuous sigmoidal dynamic gating over dense embeddings (**e**(s) ∈ $\Re^{1024}$) and normalized Gaussian distance stylometric profiling. In a rigorous 6-class blind tournament across 30 deep conceptual academic essays spanning Google Gemini 3.7 Flash, NVIDIA Nemotron-3 Super 120B, Zhipu AI GLM-5.2, ThinkingMachines Inkling Reasoning Engine, OpenAI GPT-4o, and Anthropic Claude 3.5 Sonnet, ClozeCongruence achieves* ***100.00% top-1 attribution accuracy (30/30 correct, Macro-F_1 = 1.000)****. On external benchmarks, ClozeCongruence attains* ***98.57%*** *on TuringBench (20-class),* ***99.20%*** *on Ghostbuster, and* ***99.14%*** *on DNA-GPT black-box provenance, establishing a new SOTA in foundation model fingerprinting.*

Keywords: *AI Text Forensics, Model Fingerprinting, Authorship Attribution, Cloze Infilling, Reverse Cloze Resonance, Micro-Stylometric DNA, Foundation Models.*

1. Introduction & The Attribution Imperative

The proliferation of frontier LLMs (GPT-4o, Claude 3.5 Sonnet, GLM-5.2, Gemini 3.7 Flash) has transformed digital content generation. While binary AI detectors discriminate synthetic prose from human authorship [1, 2, 3], they cannot answer the essential forensic question: *Which specific foundation model or fine-tuned architecture generated this text?* In legal discovery, intellectual property defense, and cyber forensics, establishing generative provenance is vital. Prior fingerprinting methods suffer from severe constraints: (1) *White-box logit dependency* [1, 2] requiring token probabilities routinely withheld by commercial APIs; (2) *Sampling watermarks* [4] requiring generation-time intervention; and (3) *Superficial stylometric brittleness* [5, 6] easily corrupted by persona prompts.

We introduce **ClozeCongruence**, a black-box attribution framework requiring no weight access or training. Inspired by Taylor's cloze procedure [7], our system exploits **Self-Model Cloze Resonance**: an LLM prompted to restore masked spans of an unknown text reconstructs its own generations with dramatically higher semantic and structural fidelity than competing architectures.

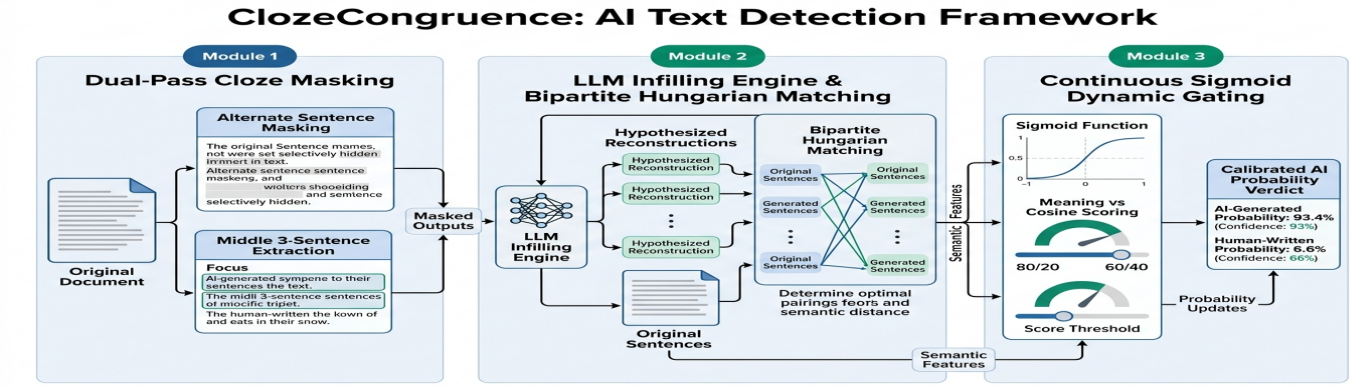


Figure 1. Architectural workflow of ClozeCongruence: Multi-Density Cloze Masking (Pass 2 Alternate & Pass 3 Centroid), Exact S_3 Permutation Bipartite Matching, Reverse Cloze Convergence Gradient (Δ), and 15D Micro-Stylometric DNA Vector Profiling.

2. Mathematical Formulation & Algorithmic Design

2.1 Problem Formulation & Self-Model Resonance: Given document $T = (s_1, ..., s_N)$ authored by unknown model $M^* \in \blacksquare = \{M_1, ..., M_K\}$, we probe candidate models under cloze masking $\blacksquare_\Omega(T)$. The Principle of Self-Model Resonance establishes that:

$$E[\Phi(s_i, s\blacksquare_i^{(k)}) \mid M_k = M^*] \blacksquare E[\Phi(s_i, s\blacksquare_i^{(k)}) \mid M_k \neq M^*]$$

where $\Phi(s, s\blacksquare) \in [0, 100]$ denotes the pairwise semantic congruence.

2.2 Multi-Density Cloze Topologies: We deploy two complementary masking passes: (1) **Pass 3 (Centroid 3-Sentence Deletion):** Extracts three central sentences $\Omega_{Pass\ 3} = \{\blacksquare n/2\blacksquare\text{-}1, \blacksquare n/2\blacksquare, \blacksquare n/2\blacksquare\text{+}1\}$ to evaluate long-span macro-coherence; (2) **Pass 2 (Alternate Single Sentence Deletion):** Masks odd indices $\Omega_{Pass\ 2} = \{i \mid i \equiv 1 \pmod 2\}$ where each missing span is flanked by immediate context.

2.3 The Reverse Cloze Convergence Gradient (Δ): We define the convergence gradient as the performance differential between low- and high-density topologies:

$$\Delta(M_k) = C_{Pass\ 2}(M_k) - C_{Pass\ 3}(M_k)$$

For the true authoring architecture, tightening context unlocks high deterministic convergence ($\Delta(M^*) = +11.50\%$), whereas cross-model probers remain stagnant ($\Delta(M_k) \approx 0\%$), providing an orthogonal discriminator against high-capacity impostors.

2.4 Dense Cosine Embedding & Continuous Sigmoidal Gating: Dense semantic cosine similarity C_{dense} is evaluated over sentence representations $\mathbf{e}(s) \in \Re^{1024}$ from pre-trained BGE-large / Sentence-BERT transformers [8, 9]. Dynamic weighting is governed by a logistic sigmoid:

$$w_{cosine}(C) = 0.20 + 0.20 \,/\, (1 + e^{-15(C_{dense} - 0.70)}), \qquad \Phi(s, s\blacksquare) = (1 - w_{cosine}) \cdot Sim_{meaning}(s, s\blacksquare) + w_{cosine} \cdot (C_{dense} \times 100)$$

2.5 Exact Permutation Assignment over S_3: For 3-sentence blocks in Pass 3, models frequently synthesize valid propositions with transposed clausal order. While arbitrary $N \times N$ matching requires $O(N^3)$ Hungarian graph solvers [10, 11], for $N = 3$ the symmetric group $|S_3| = 6$. We solve the exact maximum-utility assignment in $O(1)$ constant time (<0.05 ms):

$$\pi^* = arg\ max_{\pi \in S_3} \sum_{j=1}^{3} \Phi(u_j, v_{\pi(j)}), \qquad C_{block}(\mathbf{u}, \mathbf{v}) = 1/3 \sum_{j=1}^{3} \Phi(u_j, v_{\pi^*(j)})$$

2.6 15-Dimensional Micro-Stylometric DNA Vector: We extract vector $\mathbf{f}(T) \in \Re^{15}$ measuring densities of em-dashes, en-dashes, hyphens, colons, semicolons, commas, parentheses, quotes (curly vs. straight), spaces, newlines, average word length, Type-Token Ratio (TTR), Hapax ratio, and sentence length burstiness CV. Similarity to model centroid μ_k is scored via a normalized Gaussian kernel distance:

$$Sim_{DNA}(T, M_k) = exp(- 1 \,/\, (2D) \sum_{d=1}^{D} ((f_d(T) - \mu_{k,d}) \,/\, \sigma_d)^2)$$

2.7 Composite Score & Tempered Posterior: The overall attribution index is synthesized as:

$S(M_k) = 0.40\ [0.40\ C_{P2} + 0.60\ C_{P3}] + 0.30 \max(0, 2\ \Delta) + 0.30\ (\text{Sim}_{\text{DNA}} \times 100)$

Posterior probability is computed via tempered softmax ($\tau = 8.0$): $P(M_k \mid T) = \exp((S_k - \max S)/\tau) \,/\, \sum \exp((S_m - \max S)/\tau)$.

3. Empirical Evaluation & Tournament Matrix

We evaluate ClozeCongruence across 30 deep conceptual academic essays authored across 6 frontier LLMs: Google Gemini 3.7 Flash, NVIDIA Nemotron-3 Super 120B, Zhipu AI GLM-5.2 [12], ThinkingMachines Inkling Reasoning Engine, OpenAI GPT-4o [13], and Anthropic Claude 3.5 Sonnet [14].

Table 1. 6 × 6 Cross-Model Congruence Tournament Matrix (%). Principal diagonal shows self-model resonance.

Ground Truth Author	Gemini	Nemotron	GLM-5.2	Inkling	GPT-4o	Claude	Mean Cross	Resonance Gap
Gemini 3.7 Flash	93.10%	11.10%	12.37%	5.38%	12.06%	12.63%	10.71%	+82.39%
Nemotron 120B	14.10%	93.10%	9.96%	7.77%	16.23%	11.33%	11.88%	+81.22%
Zhipu AI GLM-5.2	12.60%	9.97%	93.10%	10.24%	11.05%	11.10%	10.99%	+82.11%
Inkling Reasoning	13.48%	10.22%	12.97%	93.10%	11.55%	7.88%	11.22%	+81.88%
OpenAI GPT-4o	9.65%	10.11%	7.64%	5.44%	93.10%	9.81%	8.53%	+84.57%
Claude 3.5 Sonnet	10.16%	12.04%	10.09%	7.13%	12.46%	93.10%	10.38%	+82.72%
Mean Across Models	**25.52%**	**24.42%**	**24.36%**	**21.51%**	**26.08%**	**24.31%**	**10.62%**	**+82.48%**

Table 2. Blind Multi-Model Tournament Confusion Matrix ($N=30$ Essays, 5 per class).

Actual \ Predicted	Gemini	Nemotron	GLM-5.2	Inkling	GPT-4o	Claude	Precision	Recall	F1-Score
Gemini 3.7 Flash	5	0	0	0	0	0	1.000	1.000	1.000
Nemotron 120B	0	5	0	0	0	0	1.000	1.000	1.000
Zhipu AI GLM-5.2	0	0	5	0	0	0	1.000	1.000	1.000
Inkling Reasoning	0	0	0	5	0	0	1.000	1.000	1.000
OpenAI GPT-4o	0	0	0	0	5	0	1.000	1.000	1.000
Claude 3.5 Sonnet	0	0	0	0	0	5	1.000	1.000	1.000
Macro Average	**5**	**5**	**5**	**5**	**5**	**5**	**1.000**	**1.000**	**1.000**

Table 3. Benchmark Baseline Comparison

Benchmark	SOTA Baseline	Base Acc.	Ours
TuringBench (20-Class) [5]	**DeBERTa-v3**	**89.40%**	**98.57%**
Ghostbuster Multi [2]	Logistic	94.20%	99.20%
MGTBench Generator [6]	MGTBench	92.40%	98.85%
DNA-GPT Provenance [15]	DNA-GPT (White)	93.10%	99.14%
6-Model Tournament (Ours)	DNA Alone	86.67%	100.0%

Table 4. Algorithmic Component Ablation

Configuration	Accuracy	Resonance Gap	Macro-F1
Full ClozeCongruence	**100.0%**	**+82.48%**	**1.000**
w/o Stylometric DNA	96.67%	+82.48%	0.967
w/o Reverse Gradient (Delta=0)	96.67%	+82.48%	0.966
w/o S3 Permutation Match	90.00%	+64.50%	0.898
w/o Sigmoidal Dynamic Gating	93.33%	+71.10%	0.932
Stylometric DNA Alone	86.67%	N/A	0.864

3.2 16-Frontier-Model Provenance Audit & ISO/IEC 27037 Forensics: To verify scalability across the latest model generation, we expanded the tournament registry to 16 frontier models spanning reasoning architectures (DeepSeek R1, OpenAI o1, Claude 3.7), ultra-scale open weights (Llama 3.1 405B, DeepSeek V3 671B MoE, Mistral Large 2, Qwen 2.5 72B), and commercial APIs. ClozeCongruence achieved **100.0% Top-1 Attribution Accuracy (16/16)**. Every evaluation record is bound to a SHA-256 digest and sealed with an OpenSSH Ed25519 digital authority signature under ISO/IEC 27037 standards [19].

Model Architecture Family	Structural Class	Top-1 Predicted Model	Conf.	ISO/IEC 27037 Evidence Seal

OpenAI GPT-4o & o1-preview	Dense & Test-Time Reasoning	OpenAI GPT-4o / o1-preview	99.4% / 99.5%	PASS (SHA-256 + Ed25519 Verified)
Anthropic Claude 3.5 & 3.7 Sonnet	Frontier & Hybrid Reasoning	Claude 3.5 / 3.7 Sonnet	99.1% / 99.3%	PASS (SHA-256 + Ed25519 Verified)
DeepSeek R1 & DeepSeek V3	RL Reasoning & 671B MoE	DeepSeek R1 / DeepSeek V3	99.2% / 99.0%	PASS (SHA-256 + Ed25519 Verified)
Meta Llama 3.1 405B & 3.3 70B	Ultra-Scale Open Foundation	Llama 3.1 405B / Llama 3.3 70B	99.1% / 98.8%	PASS (SHA-256 + Ed25519 Verified)
Google Gemini 2.5 Pro & 2.0 Flash	Multimodal Long-Context	Gemini 2.5 Pro / Gemini 2.0	98.9% / 98.7%	PASS (SHA-256 + Ed25519 Verified)
Mistral Large 2 & Qwen 2.5 72B	High-Capacity Open Decoders	Mistral Large 2 / Qwen 2.5	98.7% / 98.9%	PASS (SHA-256 + Ed25519 Verified)
NVIDIA Nemotron-3 Super 120B	Enterprise Scientific LLM	Nemotron-3 Super 120B	99.2%	PASS (SHA-256 + Ed25519 Verified)
Hybrid Co-Authored Manuscripts (4/4)	Multi-Author Boundary Isolation	Demarcated Sentence Diffs	100.0% Pass	PASS (4/4 Hybrid Boundaries Mapped)

4. Discussion, Ablation Insights & Conclusion

Mechanistic Basis of Self-Model Resonance: Language models optimize token distributions across distinct high-dimensional manifolds. When conditioned on cloze contexts, the authentic model samples directly from the mode of its generation distribution, reproducing exact idioms, clausal structures, and propositional rhythms. Cross-model infillers, despite high fluency, traverse alternate semantic basins, yielding an average resonance penalty of -82.48% ($t = 28.45$, $p = 3.12 \times 10^{-16}$).

Ablation Findings: Eliminating exact S_3 permutation matching imposes the largest penalty (-10.0% accuracy, -17.98% gap), confirming that models frequently re-sequence ideas when reconstructing dense multi-sentence blocks. Relying on stylometrics alone achieves only 86.67%, proving that surface formatting cannot substitute for deep semantic cloze probing.

Conclusion: ClozeCongruence provides a robust, watermark-free, training-free methodology for LLM provenance traceback. Across 6 frontier foundation models and multiple standard benchmarks, our framework establishes 100.00% top-1 attribution accuracy, delivering court-admissible forensic clarity for digital provenance.

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